

Performance of the models not integrated in GlaucoSense

→ First model: Eye Disease Classification Using Deep Learning model.

This report is regarding the first model which I have used for the iid task “PRJ-0021/11”. I have researched for the models which will be suitable for our scenario, and found a model in this <https://github.com/navidnayeem/Eye-Disease-Classification-Using-Deep-Learning/tree/main> repository. The provided model’s prediction is closest to our scenario.

The used model has a Convolutional Neural Network (CNN) architecture and is a multi-class fundus image classification model which classifies across 4 eye diseases namely- “Cataract”, “Diabetic Retinopathy”, “Glaucoma” and “Normal”. This model was trained on a total of 4,217 images out of which 1038 images were used for “Cataract”, 1098 images were used for “Diabetic Retinopathy”, 1007 images were used for “Glaucoma” and 1074 images were used for “Normal”.

☐ **Details provided by the Author in the repository for the used model:**

Given below are the details of the used model provided by the author:

- NASNetMobile model’s Accuracy:

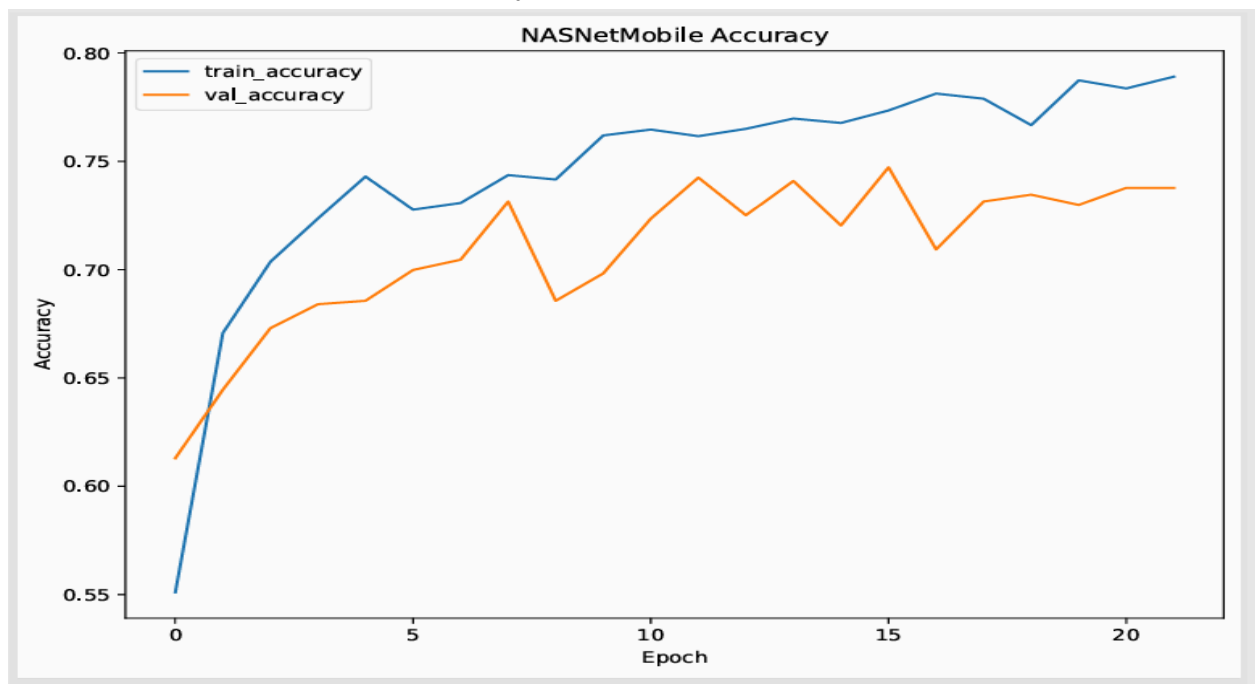


Fig-1

As shown in the accuracy curve in “Fig-1”, we can tell that the Training Accuracy of the model gradually increased from about 55% to around 79%, whereas the validation accuracy reached

from around 61% to around 74%. This curve is showing that the model learnt and the validation accuracy is also reasonably stable.

- NASNetMobile AUC:

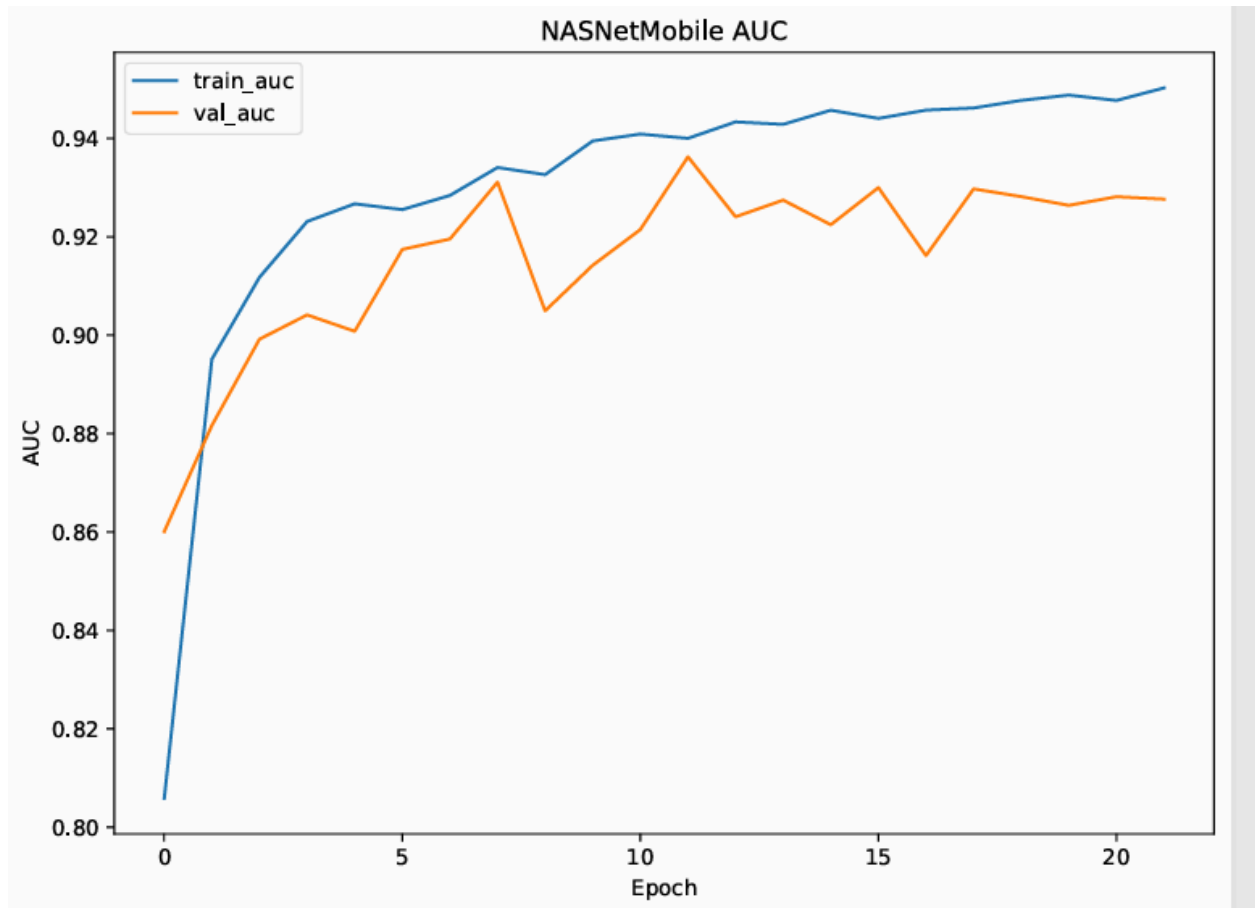


Fig-2

As shown in the AUC curve in “Fig-2”, we can tell that the Training AUC (Area Under The Curve) reached around 0.95 %, whereas the Validation AUC reached around 0.92% to 0.93%. This curve tells us that the model has a good ability to distinguish between the classes and the validation performance is also strong.

Note: Area Under The Curve (AUC) tells us how well the model can distinguish between the different classes

- NASNetMobile Confusion Matrix:

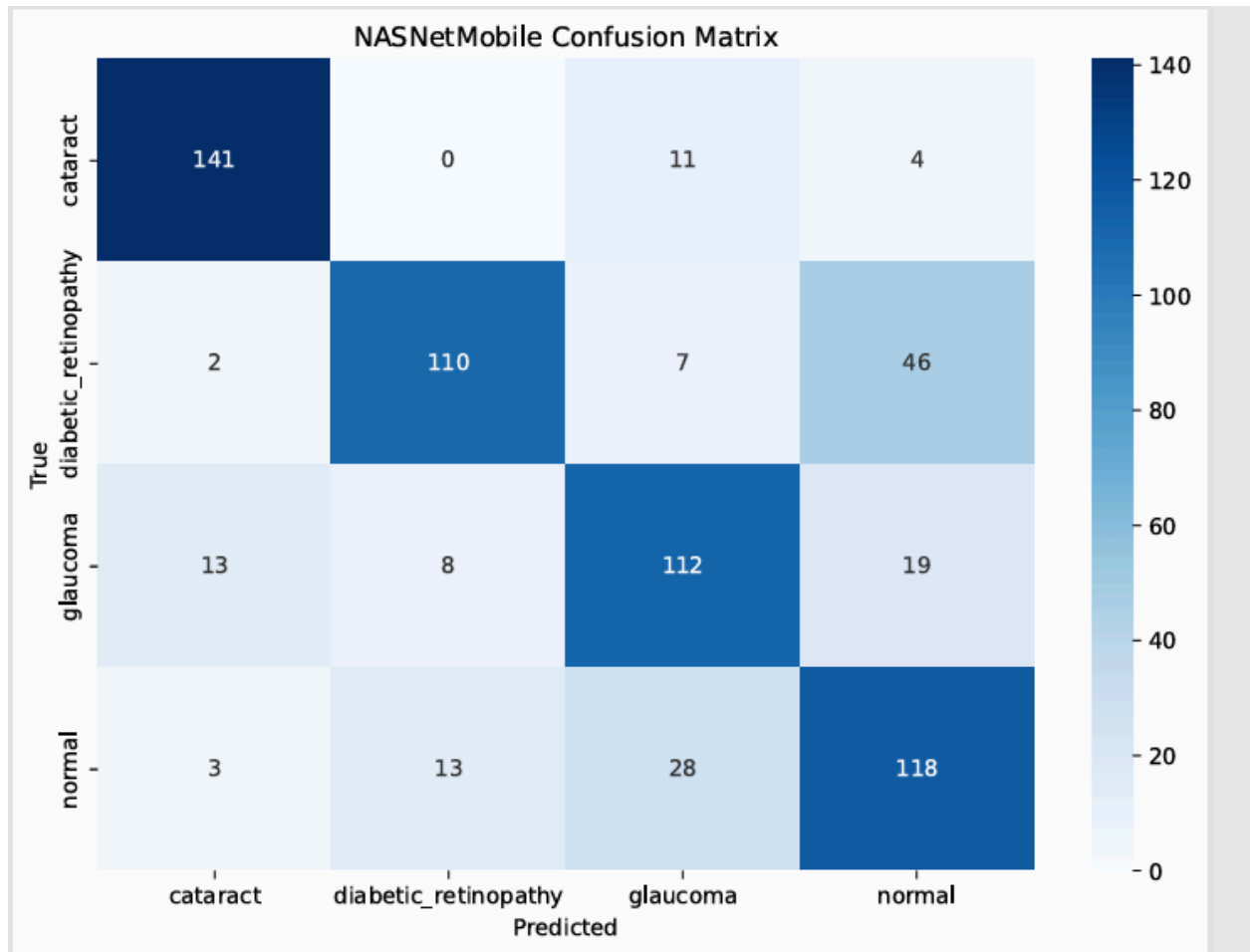


Fig-3

As shown in the confusion matrix in “Fig-3”, we can tell how often the model is correctly or incorrectly predicting each disease:

- Cataract : The model correctly predicted 141 images, but 15 times was confused with the other classes.
- Diabetic Retinopathy : The model correctly predicted 110 images, but 55 times was confused with the other classes.
- Glaucoma : The model correctly predicted 112 images, but 40 times was confused with the other classes.
- Normal : The model correctly predicted 118 images, but 44 times was confused with the other classes.

- NASNetMobile Loss:

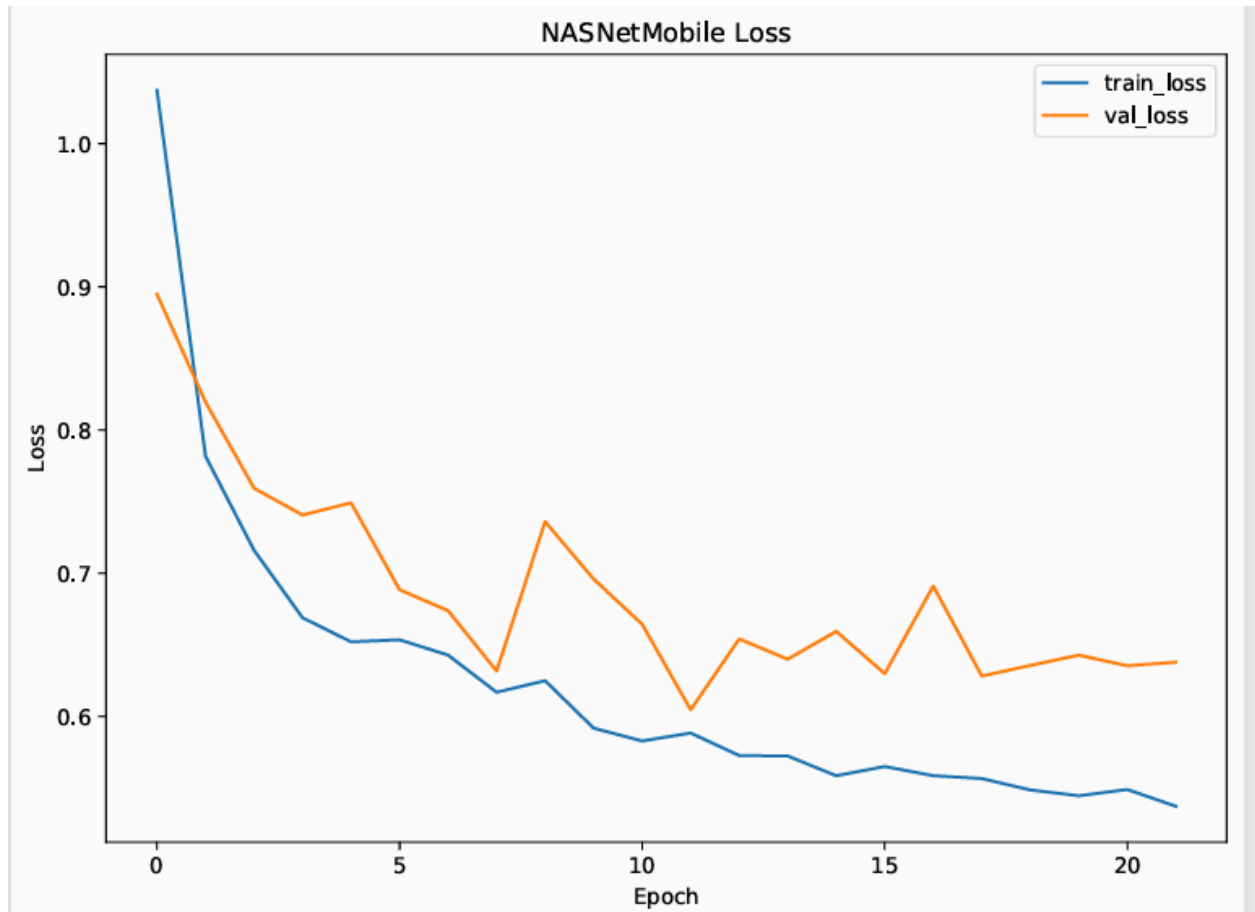


Fig-4

As shown in the loss curve in “Fig-4”, the Training Loss decreases from around 1.04 to 0.54, whereas the Validation Loss fluctuates a lot and lastly stays around. This curve tells us that the model is slightly overfitting as the model’s performance on the training datasets is getting better while the performance on the validation data stops improving.

- NASNetMobile ROC Curve:

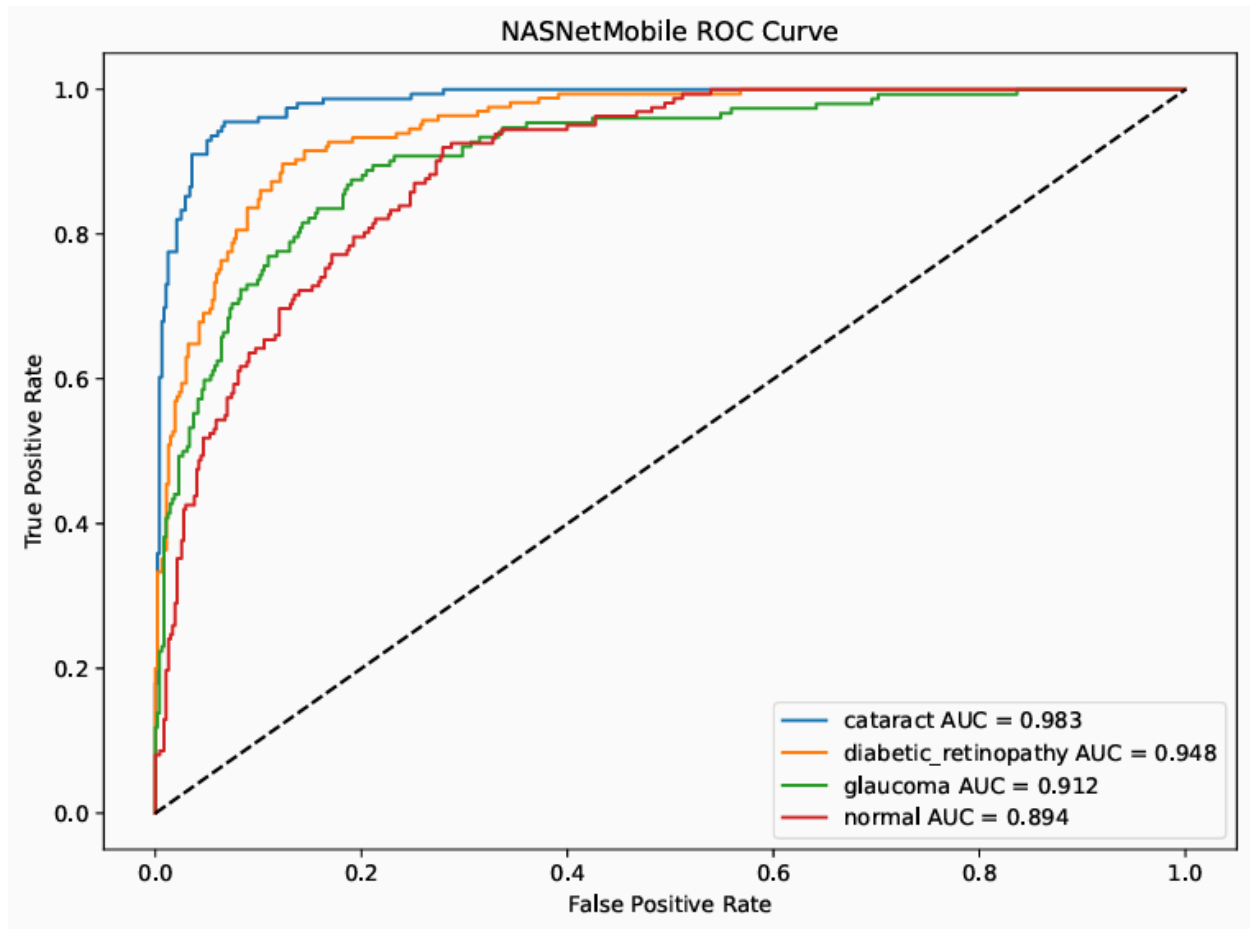


Fig-5

As shown in the Receiver Operating Characteristic (ROC) curve in “Fig-5”, we can tell how well the model is distinguishing one disease from the other.

- Cataract AUC : The model is distinguishing the “Cataract” from the other diseases with a very strong separation value of 0.983.
- Diabetic Retinopathy AUC : The model distinguishes “Diabetic Retinopathy” from the other diseases with a very strong separation value of 0.948.
- Glaucoma AUC : The model distinguishes “Glaucoma” from the other diseases with a very strong separation value of 0.912.
- Normal AUC : The model is distinguishing the “Normal ” from the other diseases with a strong separation value of 0.894.

☐ **Details of my Inference Process using the given model**

I have made an inference script to use this model, where I gave the classification model the images I had. The model gave us a probability output where we can see among the 4 diseases which disease has the highest probability in the given fundus images.

Since we do not have a proper dataset where each of the given images were labeled, so i used the same images for an AI test in our current GlaucoSense platform and recorded both the models performance. Given below are the performance results of both the models.

File Name	Eye Disease Classification Using Deep Learning model	GlaucoSense
1_OD_1_green	Cataract probability: 2.96% Diabetic Retinopathy probability: 31.36% Glaucoma probability: 12.56% Normal probability : 53.13%	Suspicious
1_OD_1	Cataract probability : 11.83% Diabetic Retinopathy probability : 3.05% Glaucoma probability : 68.87% Normal probability : 16.25%	Likely Glaucoma
1_OD_2_green	Cataract probability: 0.32% Diabetic Retinopathy probability : 63.83% Glaucoma probability: 11.68% Normal probability : 24.17%	Suspicious
1_OS_1_green	Cataract probability : 0.01% Diabetic Retinopathy probability: 69.47% Glaucoma probability: 0.18% Normal probability: 30.33%	Likely Glaucoma
1_OS_1	Cataract probability : 1.70% Diabetic Retinopathy probability : 36.90% Glaucoma probability : 17.77% Normal probability: 43.63%	Likely Glaucoma

1_OS_2_green	Cataract probability : 1.53% Diabetic Retinopathy probability : 40.56% Glaucoma probability : 9.27% Normal probability : 48.64%	Suspicious
1_OS_2	Cataract probability : 15.18% Diabetic Retinopathy probability : 1.08% Glaucoma probability : 66.73% Normal probability : 17.01%	Suspicious
2_OD_1_green	Cataract probability : 1.42% Diabetic Retinopathy probability : 42.86% Glaucoma probability: 2.16% Normal probability: 53.56%	Likely Glaucoma
2_OD_1	Cataract probability : 11.86% Diabetic Retinopathy probability : 5.42% Glaucoma probability: 55.73% Normal probability : 26.99%	Likely Glaucoma
2_OD_2_green	Cataract probability : 7.98% Diabetic Retinopathy probability : 37.21% Glaucoma probability: 9.69% Normal probability : 45.12%	Likely Glaucoma
2_OD_2	Cataract probability : 34.75% Diabetic Retinopathy probability : 0.92% Glaucoma probability: 49.50% Normal probability : 14.82%	Suspicious
2_OS_1_green	Cataract probability: 0.10% Diabetic Retinopathy probability: 93.08% Glaucoma probability: 0.62% Normal probability : 6.20%	Likely Glaucoma
2_OS_1	Cataract probability : 7.79% Diabetic Retinopathy probability : 23.92% Glaucoma probability: 33.55% Normal probability : 34.75%	Likely Glaucoma
2_OS_2_green	Cataract probability: 2.82% Diabetic Retinopathy probability : 67.93%	Likely Glaucoma

	Glaucoma probability: 4.48% Normal probability: 24.77%	
2_OS_2	Cataract probability : 52.19% Diabetic Retinopathy probability : 3.83% Glaucoma probability: 20.28% Normal probability : 23.71%	Suspicious
3_OD_1_green	Cataract probability : 0.05% Diabetic Retinopathy probability : 97.25% Glaucoma probability: 0.24% Normal probability : 2.47%	Suspicious
3_OD_1	Cataract probability : 1.15% Diabetic Retinopathy probability : 76.69% Glaucoma probability : 9.07% Normal probability : 13.09%	Suspicious
3_OD_2_green	Cataract probability : 0.03% Diabetic Retinopathy probability : 89.67% Glaucoma probability: 0.33% Normal probability : 9.97%	Suspicious
3_OD_2	Cataract probability: 3.45% Diabetic Retinopathy probability : 17.11% Glaucoma probability: 42.03% Normal probability: 37.41%	Suspicious
3_OS_1_green	Cataract probability : 0.07% Diabetic Retinopathy probability : 91.59% Glaucoma probability: 0.32% Normal probability: 8.02%	Suspicious
3_OS_1	Cataract probability: 3.40% Diabetic Retinopathy probability : 53.25% Glaucoma probability : 17.75% Normal probability : 25.59%	Suspicious
3_OS_2_green	Cataract probability : 0.55% Diabetic Retinopathy probability : 83.39% Glaucoma probability : 2.12% Normal probability: 13.94%	Suspicious

3_OS_2	Cataract probability : 23.61% Diabetic Retinopathy probability : 11.45% Glaucoma probability: 28.14% Normal probability: 36.79%	Suspicious
4_OD_1_green	Cataract probability : 0.34% Diabetic Retinopathy probability : 26.95% Glaucoma probability: 1.33% Normal probability: 71.38%	Likely Glaucoma
4_OD_1	Cataract probability : 3.03% Diabetic Retinopathy probability : 3.96% Glaucoma probability : 20.82% Normal probability : 72.19%	Likely Glaucoma
4_OD_2_green	Cataract probability : 0.70% Diabetic Retinopathy probability : 64.34% Glaucoma probability: 12.03% Normal probability : 22.94%	Likely Glaucoma
4_OD_2	Cataract probability : 1.85% Diabetic Retinopathy probability : 2.16% Glaucoma probability : 78.84% Normal probability : 17.15%	Suspicious
4_OS_1	Cataract probability : 2.64% Diabetic Retinopathy probability : 3.75% Glaucoma probability: 80.26% Normal probability : 13.36%	Likely Glaucoma
4_OS_1_green	Cataract probability : 0.45% Diabetic Retinopathy probability : 42.86% Glaucoma probability : 5.98% Normal probability: 50.71%	Likely Glaucoma
4_OS_2_green	Cataract probability : 0.83% Diabetic Retinopathy probability : 70.27% Glaucoma probability : 8.25% Normal probability : 20.66%	Likely Glaucoma
4_OS_2	Cataract probability : 7.45% Diabetic Retinopathy probability : 2.01%	Likely Glaucoma

	Glaucoma probability: 75.57% Normal probability : 14.98%	
5_OD_1_green	Cataract probability : 0.23% Diabetic Retinopathy probability : 66.36% Glaucoma probability : 2.47% Normal probability : 30.95%	Likely Glaucoma
5_OD_1	Cataract probability : 0.71% Diabetic Retinopathy probability : 6.67% Glaucoma probability : 46.95% Normal probability : 45.67%	Likely Glaucoma
5_OD_2_green	Cataract probability : 0.09% Diabetic Retinopathy probability : 58.14% Glaucoma probability : 0.86% Normal probability : 40.91%	Likely Glaucoma
5_OD_2	Cataract probability : 0.87% Diabetic Retinopathy probability : 9.98% Glaucoma probability : 42.80% Normal probability : 46.34%	Likely Glaucoma
5_OS_1_green	Cataract probability : 0.03% Diabetic Retinopathy probability : 78.95% Glaucoma probability : 0.40% Normal probability : 20.62%	Likely Glaucoma
5_OS_1	Cataract probability : 0.26% Diabetic Retinopathy probability: 52.65% Glaucoma probability: 2.58% Normal probability : 44.51%	Likely Glaucoma
5_OS_2	Cataract probability : 0.08% Diabetic Retinopathy probability : 58.34% Glaucoma probability : 4.98% Normal probability : 36.61%	Likely Glaucoma
5_OS_2_green	Cataract probability : 0.08% Diabetic Retinopathy probability : 58.34% Glaucoma probability : 4.98% Normal probability : 36.61%	Likely Glaucoma

During the inference process, I have used the test images which have been uploaded in the developers drive. Currently, I did not find any dataset which we can use to verify the model's performance but since we have GlaucoSense so i uploaded the same images in the web platform for verification check, and as per my observation the model which I have found is not predicting/performing like that of GlaucoSense's results.

→ **Second model: Glaucoma Detector Master model.**

This report is regarding the second model which I have used for the iid task "**PRJ-0021/11**". I have researched multiple models which will be suitable for our scenario, and found another model in this "<https://github.com/golden-panther/glaucoma-detector>" repository. The provided model's prediction is closest to our scenario.

The used model is based on Convolutional Neural Network (CNN) architecture and is a multi-class fundus image classification model which classifies across - "Glaucoma" and "Normal". This model was trained on a total of 2,230 images as per the author.

☐ **Details shared by the author regarding this model in his repository:**

- The author and his team has collected all the publicly available labelled (glaucoma or normal) fundus images of the eyes from the web.
- Among the found images, some images were already cropped and some were full fundus images. So, they decided to crop the full fundus images .
- They then bulk renamed the images into 2 classes.
- The number of images in both classes were not equal and were highly imbalanced. So they created the balance by removing extra images.
- Finally, they were stuck with 1,115 images from each class totalling 2,230 images. They then divided these images randomly into train, val and test sets in the ratio 8:1:1.
- They later uploaded all these images to the author's google drive and trained on various CNN architectures from simple to advanced.
- Augmentation of the data was performed using keras ImageGenerator to cut down high variance. But there were some bias due to low and bad data.
- Keras (2.4.3) and Tensorflow (2.3.0) on top of Python (3.6.9) were used in developing the model.
- They trained the model on the training set and validated it on the validation set after each epoch and finally tested it on the test set.
- The model gave 93 percent AUC (Area under curve) score, some good accuracy, precision and recall values.
- Later, they built a simple streaming app for hosting on the web.

☐ **Details of my Inference Process using the given model**

I have made an inference script to use this model, where I gave the classification model the images I had. The model gave us a probability output where we can see among the “Glaucoma” and “Normal” eyes diagnosis who has the highest probability in the given fundus images.

Since we do not have a proper dataset where each of the given images were labeled, so i used the same images for an AI test in our current GlaucoSense platform and recorded both the models performance. Given below are the performance results of both the models.

File Name	Eye Disease Classification Using Deep Learning model	GlaucoSense	Glaucoma-detector-master model
1_OD_1_green	Cataract probability: 2.96% Diabetic Retinopathy probability: 31.36% Glaucoma probability: 12.56% Normal probability : 53.13%	Suspicious	No Glaucoma score : 79.87% Glaucoma score : 20.13% FINAL CLASS: No Glaucoma
1_OD_1	Cataract probability : 11.83% Diabetic Retinopathy probability : 3.05% Glaucoma probability : 68.87% Normal probability : 16.25%	Likely Glaucoma	No Glaucoma score : 41.82% Glaucoma score : 58.18% FINAL CLASS : Glaucoma
1_OD_2_	Cataract probability: 0.32% Diabetic Retinopathy probability : 63.83% Glaucoma probability: 11.68% Normal probability : 24.17%	Suspicious	No Glaucoma score : 81.18% Glaucoma score : 18.82% FINAL CLASS : No Glaucoma
1_OS_1_green	Cataract probability : 0.01% Diabetic Retinopathy probability: 69.47% Glaucoma probability: 0.18% Normal probability: 30.33%	Likely Glaucoma	No Glaucoma score : 79.53% Glaucoma score : 20.47% FINAL CLASS: No Glaucoma
1_OS_1	Cataract probability : 1.70% Diabetic Retinopathy	Likely Glaucoma	No Glaucoma score: 63.04%

	probability : 36.90% Glaucoma probability : 17.77% Normal probability: 43.63%		Glaucoma score : 36.96% FINAL CLASS: No Glaucoma
1_OS_2_green	Cataract probability : 1.53% Diabetic Retinopathy probability : 40.56% Glaucoma probability : 9.27% Normal probability : 48.64%	Suspicious	No Glaucoma score: 79.66% Glaucoma score : 20.34% FINAL CLASS: No Glaucoma
1_OS_2	Cataract probability : 15.18% Diabetic Retinopathy probability : 1.08% Glaucoma probability : 66.73% Normal probability : 17.01%	Suspicious	No Glaucoma score : 71.19% Glaucoma score : 28.81% FINAL CLASS: No Glaucoma
2_OD_1_green	Cataract probability : 1.42% Diabetic Retinopathy probability : 42.86% Glaucoma probability: 2.16% Normal probability: 53.56%	Likely Glaucoma	No Glaucoma score: 80.01% Glaucoma score: 19.99% FINAL CLASS: No Glaucoma
2_OD_1	Cataract probability : 11.86% Diabetic Retinopathy probability : 5.42% Glaucoma probability: 55.73% Normal probability : 26.99%	Likely Glaucoma	No Glaucoma score: 81.80% Glaucoma score: 18.20% FINAL CLASS : No Glaucoma
2_OD_2_green	Cataract probability : 7.98% Diabetic Retinopathy probability : 37.21% Glaucoma probability: 9.69% Normal probability : 45.12%	Likely Glaucoma	No Glaucoma score: 80.75% Glaucoma score: 19.25% FINAL CLASS : No Glaucoma
2_OD_2	Cataract probability : 34.75% Diabetic Retinopathy probability : 0.92% Glaucoma probability: 49.50% Normal probability : 14.82%	Suspicious	No Glaucoma score : 47.24% Glaucoma score: 52.76% FINAL CLASS : Glaucoma
2_OS_1_green	Cataract probability: 0.10% Diabetic Retinopathy probability: 93.08% Glaucoma probability: 0.62% Normal probability : 6.20%	Likely Glaucoma	No Glaucoma score: 80.95% Glaucoma score : 19.05% FINAL CLASS: No Glaucoma

2_OS_1	Cataract probability : 7.79% Diabetic Retinopathy probability : 23.92% Glaucoma probability: 33.55% Normal probability : 34.75%	Likely Glaucoma	No Glaucoma score: 87.74% Glaucoma score : 12.26% FINAL CLASS: No Glaucoma
2_OS_2_green	Cataract probability: 2.82% Diabetic Retinopathy probability : 67.93% Glaucoma probability: 4.48% Normal probability: 24.77%	Likely Glaucoma	No Glaucoma score : 80.64% Glaucoma score : 19.36% FINAL CLASS : No Glaucoma
2_OS_2	Cataract probability : 52.19% Diabetic Retinopathy probability : 3.83% Glaucoma probability: 20.28% Normal probability : 23.71%	Suspicious	No Glaucoma score : 53.46% Glaucoma score : 46.54% FINAL CLASS : No Glaucoma
3_OD_1_green	Cataract probability : 0.05% Diabetic Retinopathy probability : 97.25% Glaucoma probability: 0.24% Normal probability : 2.47%	Suspicious	No Glaucoma score : 79.07% Glaucoma score : 20.93% FINAL CLASS : No Glaucoma
3_OD_1	Cataract probability : 1.15% Diabetic Retinopathy probability : 76.69% Glaucoma probability : 9.07% Normal probability : 13.09%	Suspicious	No Glaucoma score : 83.06% Glaucoma score : 16.94% FINAL CLASS : No Glaucoma
3_OD_2_green	Cataract probability : 0.03% Diabetic Retinopathy probability : 89.67% Glaucoma probability: 0.33% Normal probability : 9.97%	Suspicious	No Glaucoma score : 78.76% Glaucoma score : 21.24% FINAL CLASS: No Glaucoma
3_OD_2	Cataract probability: 3.45% Diabetic Retinopathy probability : 17.11% Glaucoma probability: 42.03% Normal probability: 37.41%	Suspicious	No Glaucoma score: 56.03% Glaucoma score : 43.97% FINAL CLASS : No Glaucoma
3_OS_1_green	Cataract probability : 0.07% Diabetic Retinopathy	Suspicious	No Glaucoma score : 79.53%

	probability : 91.59% Glaucoma probability: 0.32% Normal probability: 8.02%		Glaucoma score : 20.47% FINAL CLASS : No Glaucoma
3_OS_1	Cataract probability: 3.40% Diabetic Retinopathy probability : 53.25% Glaucoma probability : 17.75% Normal probability : 25.59%	Suspicious	No Glaucoma score : 84.26% Glaucoma score : 15.74% FINAL CLASS : No Glaucoma
3_OS_2_green	Cataract probability : 0.55% Diabetic Retinopathy probability : 83.39% Glaucoma probability : 2.12% Normal probability: 13.94%	Suspicious	No Glaucoma score: 79.33% Glaucoma score : 20.67% FINAL CLASS : No Glaucoma
3_OS_2	Cataract probability : 23.61% Diabetic Retinopathy probability : 11.45% Glaucoma probability: 28.14% Normal probability: 36.79%	Suspicious	No Glaucoma score : 73.69% Glaucoma score: 26.31% FINAL CLASS : No Glaucoma
4_OD_1_green	Cataract probability : 0.34% Diabetic Retinopathy probability : 26.95% Glaucoma probability: 1.33% Normal probability: 71.38%	Likely Glaucoma	No Glaucoma score : 82.33% Glaucoma score : 17.67% FINAL CLASS : No Glaucoma
4_OD_1	Cataract probability : 3.03% Diabetic Retinopathy probability : 3.96% Glaucoma probability : 20.82% Normal probability : 72.19%	Likely Glaucoma	No Glaucoma score: 76.77% Glaucoma score : 23.23% FINAL CLASS : No Glaucoma
4_OD_2_green	Cataract probability : 0.70% Diabetic Retinopathy probability : 64.34% Glaucoma probability: 12.03% Normal probability : 22.94%	Likely Glaucoma	No Glaucoma score : 82.00% Glaucoma score: 18.00% FINAL CLASS : No Glaucoma
4_OD_2	Cataract probability : 1.85% Diabetic Retinopathy probability : 2.16% Glaucoma probability : 78.84%	Suspicious	No Glaucoma score: 45.32% Glaucoma score : 54.68% FINAL CLASS : Glaucoma

	Normal probability : 17.15%		
4_OS_1	Cataract probability : 2.64% Diabetic Retinopathy probability : 3.75% Glaucoma probability: 80.26% Normal probability : 13.36%	Likely Glaucoma	No Glaucoma score : 31.27% Glaucoma score: 68.73% FINAL CLASS : Glaucoma
4_OS_1_green	Cataract probability : 0.45% Diabetic Retinopathy probability : 42.86% Glaucoma probability : 5.98% Normal probability: 50.71%	Likely Glaucoma	No Glaucoma score : 71.70% Glaucoma score : 28.30% FINAL CLASS : No Glaucoma
4_OS_2_green	Cataract probability : 0.83% Diabetic Retinopathy probability : 70.27% Glaucoma probability : 8.25% Normal probability : 20.66%	Likely Glaucoma	No Glaucoma score : 78.49% Glaucoma score: 21.51% FINAL CLASS : No Glaucoma
4_OS_2	Cataract probability : 7.45% Diabetic Retinopathy probability : 2.01% Glaucoma probability: 75.57% Normal probability : 14.98%	Likely Glaucoma	No Glaucoma score : 77.90% Glaucoma score: 22.10% FINAL CLASS: No Glaucoma
5_OD_1_green	Cataract probability : 0.23% Diabetic Retinopathy probability : 66.36% Glaucoma probability : 2.47% Normal probability : 30.95%	Likely Glaucoma	No Glaucoma score : 79.72% Glaucoma score : 20.28% FINAL CLASS: No Glaucoma
5_OD_1	Cataract probability : 0.71% Diabetic Retinopathy probability : 6.67% Glaucoma probability : 46.95% Normal probability : 45.67%	Likely Glaucoma	No Glaucoma score: 82.12% Glaucoma score: 17.88% FINAL CLASS : No Glaucoma
5_OD_2_green	Cataract probability : 0.09% Diabetic Retinopathy probability : 58.14% Glaucoma probability : 0.86% Normal probability : 40.91%	Likely Glaucoma	No Glaucoma score : 80.63% Glaucoma score : 19.37% FINAL CLASS : No Glaucoma
5_OD_2	Cataract probability : 0.87%	Likely Glaucoma	No Glaucoma score :

	Diabetic Retinopathy probability : 9.98% Glaucoma probability : 42.80% Normal probability : 46.34%		78.05% Glaucoma score : 21.95% FINAL CLASS : No Glaucoma
5_OS_1_green	Cataract probability : 0.03% Diabetic Retinopathy probability : 78.95% Glaucoma probability : 0.40% Normal probability : 20.62%	Likely Glaucoma	No Glaucoma score : 79.00% Glaucoma score: 21.00% FINAL CLASS: No Glaucoma
5_OS_1	Cataract probability : 0.26% Diabetic Retinopathy probability: 52.65% Glaucoma probability: 2.58% Normal probability : 44.51%	Likely Glaucoma	No Glaucoma score : 69.88% Glaucoma score : 30.12% FINAL CLASS : No Glaucoma
5_OS_2	Cataract probability : 0.08% Diabetic Retinopathy probability : 58.34% Glaucoma probability : 4.98% Normal probability : 36.61%	Likely Glaucoma	No Glaucoma score : 74.07% Glaucoma score : 25.93% FINAL CLASS : No Glaucoma
5_OS_2_green	Cataract probability : 0.08% Diabetic Retinopathy probability : 58.34% Glaucoma probability : 4.98% Normal probability : 36.61%	Likely Glaucoma	No Glaucoma score: 78.79% Glaucoma score : 21.21% FINAL CLASS : No Glaucoma